

1. Create VPC with 2 private and 2 public subnets.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
training-vpc

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 [Info](#) 65,536 IP

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)
Default

► **Encryption settings - optional**

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

Preview

VPC [Show details](#)
Your AWS virtual network

Subnets (4)
Subnets within this VPC

us-east-1a
training-vpc-subnet-public1-us-east-1a
training-vpc-subnet-private1-us-east-1a
us-east-1b
training-vpc-subnet-public2-us-east-1b
training-vpc-subnet-private2-us-east-1b

Route table

Virtual private cloud					
Your VPC					
Subnets					
Route tables					
Internet gateways					
Egress-only internet gateways					
Carrier gateways					
DHCP option sets					
Elastic IPs					
Managed prefix lists					
NAT gateways					
Peering connections					
Route servers					
Security					
Network ACLs					
Security groups					
PrivateLink and Lattice					
Getting started					

☐ training-vpc-subnet-public2-us-east-1b	subnet-0415d5381f19059d1	Available	vpc-02848492a89789f6a train...	☐ OFF	10.0
☐ training-vpc-subnet-private1-us-east-1a	subnet-0430e45dbbdc74d78	Available	vpc-02848492a89789f6a train...	☐ OFF	10.0
☐ training-vpc-subnet-public1-us-east-1a	subnet-039b798918a4f16f2	Available	vpc-02848492a89789f6a train...	☐ OFF	10.0
☐ training-vpc-subnet-private2-us-east-1b	subnet-0a748a200f8c3e0a	Available	vpc-02848492a89789f6a train...	☐ OFF	10.0

subnet-0f90d36f06ddc80b1

[Details](#) [Flow logs](#) [Route table](#) [Network ACL](#) [CIDR reservations](#) [Sharing](#) [Tags](#) [Related resources](#)

Details

Subnet ID subnet-0f90d36f06ddc80b1	Subnet ARN arn:aws:ec2:us-east-1:231161110641:subnet/subnet-0f90d36f06ddc80b1	State Available	Block Public Access ☐ OFF
IPv4 CIDR 172.31.100.0/24	Available IPv4 addresses 16,777,216	IPv6 CIDR -	IPv6 CIDR association ID -

2. Enable DNS Hostname in VPC.

Edit VPC settings [Info](#)

VPC details

VPC ID
vpc-02848492a89789f6a

Name
training-vpc-vpc [EditVpcSettings](#)

DHCP settings

DHCP option set [Info](#)
dopt-043a4f3ac4086fee2

DNS settings

☒ Enable DNS resolution [Info](#)

☒ Enable DNS hostnames [Info](#)

Network Address Usage metrics settings

☐ Enable Network Address Usage metrics [Info](#)

[Cancel](#) [Save](#)

3. Enable Auto Assign Public IP in 2 public subnets.

VPC > Subnets > subnet-039b798918a5f16f2 > Edit subnet settings

Subnet

Subnet ID
subnet-039b798918a5f16f2

Name
training-vpc-subnet-public1-us-east-1a

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

Enable auto-assign public IPv4 address

Enable auto-assign customer-owned IPv4 address

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

Enable resource name DNS A record on launch

Enable resource name DNS AAAA record on launch

Hostname type

Resource name

IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

Enable DNS64

VPC > Subnets > subnet-0414d3381f19059d1 > Edit subnet settings

Subnet

Subnet ID
subnet-0414d3381f19059d1

Name
training-vpc-subnet-public2-us-east-1b

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

Enable auto-assign public IPv4 address

Enable auto-assign customer-owned IPv4 address

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

Enable resource name DNS A record on launch

Enable resource name DNS AAAA record on launch

Hostname type

Resource name

IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

Enable DNS64

4. Add 2 private subnets in private route table.

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Route tables (1/4)

Last updated 2 minutes ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	training-vpc-rtb-private2-us-east-1b	rtb-0297176d74b88059a	subnet-0a748a200df6d3...	-	No	vpc-02848492a897...
☐	training-vpc-rtb-public	rtb-093aef62d0d9037ba	-	-	No	vpc-02848492a897...
☒	training-vpc-rtb-private1-us-east-1a	rtb-096082a3666a1e02d	subnet-0d30e4dbbb4e74...	-	No	vpc-02848492a897...
☐	-	rtb-08b39bcf8c030c99a	-	-	Yes	vpc-02848492a897...

rtb-096082a3666a1e02d / training-vpc-rtb-private1-us-east-1a

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Details

Route table ID

rtb-096082a3666a1e02d

VPC

vpc-02848492a89789f6a | training-vpc-

Main

No

Owner ID

231161110641

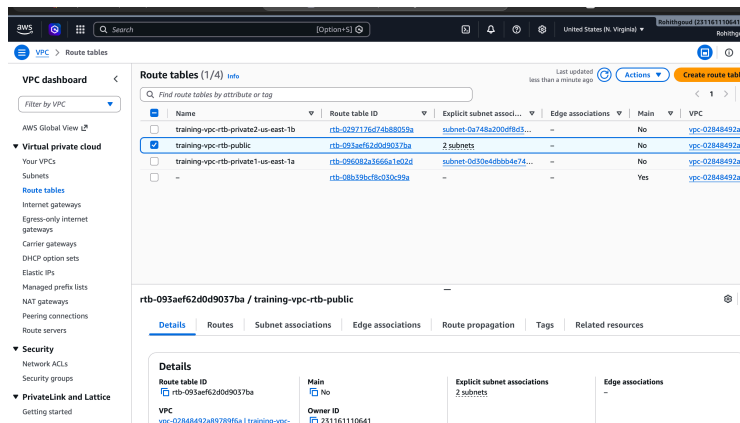
Explicit subnet associations

subnet-0d30e4dbbb4e74d78 / training-vpc-subnet-private1-us-east-1a

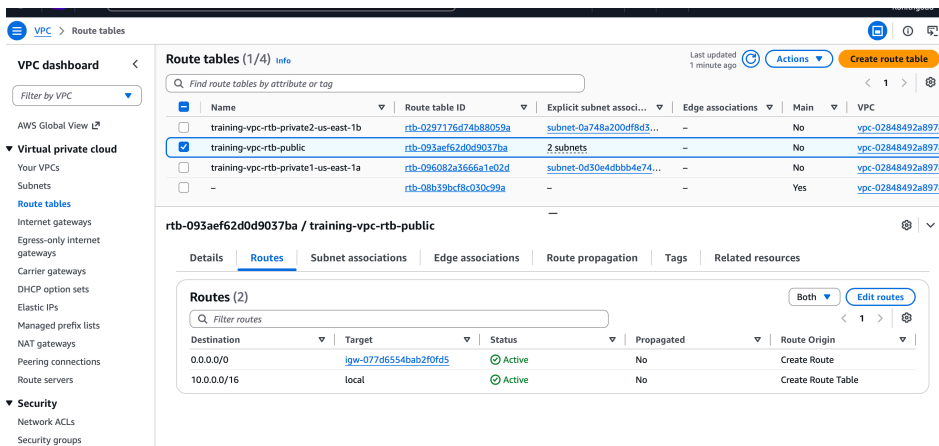
Edge associations

-

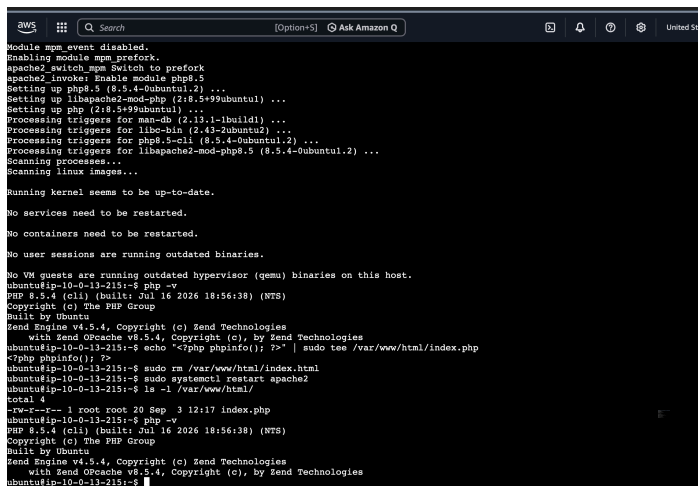
5. Add 2 public subnets in public route table.

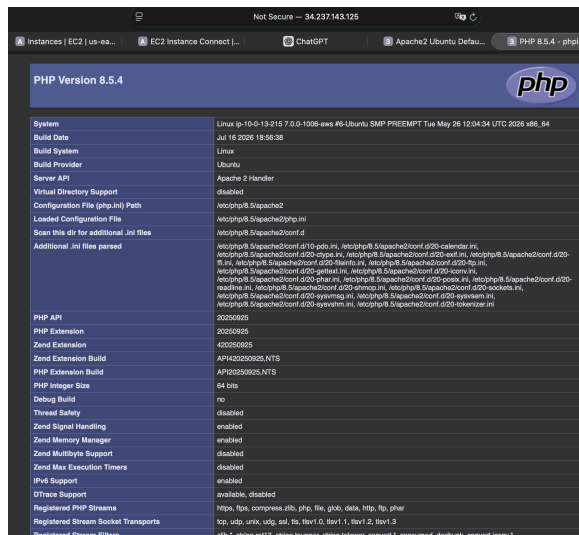


6. Public route table will have the routes to internet and local.



7. Create EC2 in public subnet with t2.micro and install PHP.





8. Configure NAT gateway in public subnet and connect to private instance.

VPC > NAT gateways > nat-1abbaf5d7abca730d

NAT gateway nat-1abbaf5d7abca730d | my-nat-gateway was created successfully.

nat-1abbaf5d7abca730d / my-nat-gateway

Details

NAT gateway ID nat-1abbaf5d7abca730d	Availability mode Regional	State Pending	State message -
NAT gateway ARN arn:aws:ec2:us-east-1:231161110641:natgateway/nat-1abbaf5d7abca730d	Connectivity type Public	Created Thursday, September 3, 2026 at 18:14:46 GMT+5:30	Deleted -
VPC vpc-02848492a89789f6a / training-vpc-vpc			
Method of EIP allocation Automatic			

IP addresses | Monitoring | Flow logs | Tags

Associated IP addresses

No associated IP addresses found. If this NAT gateway was recently created with the automatic allocation method, your IP addresses are still being allocated. Click the refresh button to view them.

VPC > Route tables > rtb-096082a3666a1e02d

Updated routes for rtb-096082a3666a1e02d / training-vpc-rtb-private1-us-east-1a successfully

rtb-096082a3666a1e02d / training-vpc-rtb-private1-us-east-1a

Details

Route table ID rtb-096082a3666a1e02d	Main No	Explicit subnet associations subnet-0d30e4dbbb4e74d78 / training-vpc-subnet-private1-us-east-1a	Edge associations -
VPC vpc-02848492a89789f6a / training-vpc-vpc	Owner ID 231161110641		

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1abbaf5d7abca730d	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

9. Install Apache Tomcat in private EC2 and deploy a sample app.

```

root@ip-10-0-131-33 ~# dnf install java-17-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository                               533 kB/s | 69 kB   00:00
dependencies resolved.

=====
Package                                Architecture      Version           Repository        Size
Installing:
java-17-amazon-corretto                x86_64            1:17.0.20+10-l.amzn2023.1  amazonlinux      220 k
Installing dependencies:
alsa-lib                                x86_64            1.2.7.2-1.amzn2023.0.4    amazonlinux      503 k
cairo                                    x86_64            1.18.0-4.amzn2023.0.2    amazonlinux      717 k
dejavu-sans-fonts                       noarch            2.37-16.amzn2023.0.2     amazonlinux      1.3 M
dejavu-sans-mono-fonts                  noarch            2.37-16.amzn2023.0.2     amazonlinux      467 k
dejavu-serif-fonts                      noarch            2.37-16.amzn2023.0.2     amazonlinux      1.0 M
fontconfig                              x86_64            2.13.94-2.amzn2023.0.2   amazonlinux      273 k
fonts-filesystem                        noarch            1:2.0.5-12.amzn2023.0.2  amazonlinux      9.5 k
freetype                                x86_64            2.13.2-5.amzn2023.0.2    amazonlinux      422 k
glib3                                    x86_64            5.2.1-9.amzn2023.0.4     amazonlinux      49 k
google-noto-fonts-common                noarch            20240401-1.amzn2023.0.2  amazonlinux      17 k

```

10. Configure VPC flow logs and store the logs in S3 and CloudWatch.

